

**Tribhuvan University
Amrit Campus**



Lab Report - SPM

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Lab 1: Draw the precedence network diagram and Gantt Chart using MS Project.

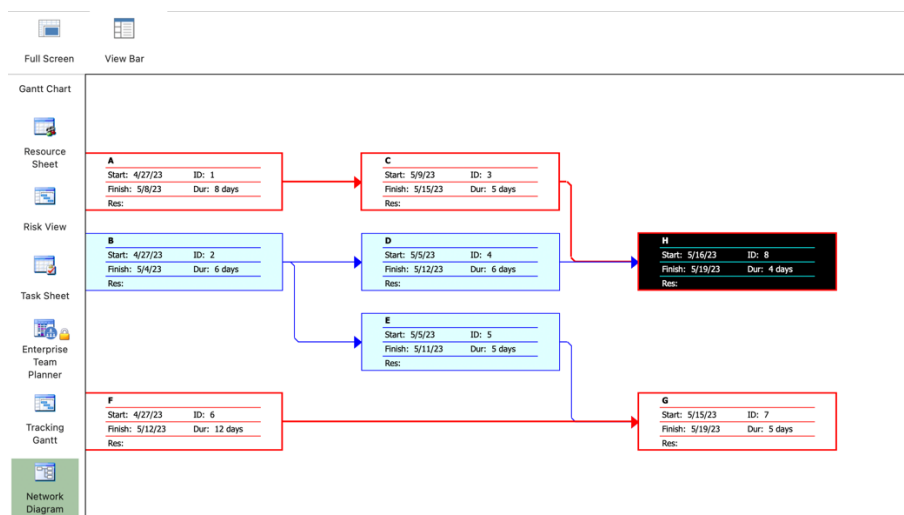
(Consider the following table of information:)

Activity	Duration(week)	Precedents
A	8	None
B	6	None
C	5	A
D	6	B
E	5	B
F	12	None
G	5	E,F
H	4	C,D

Solution:

Network Diagram:

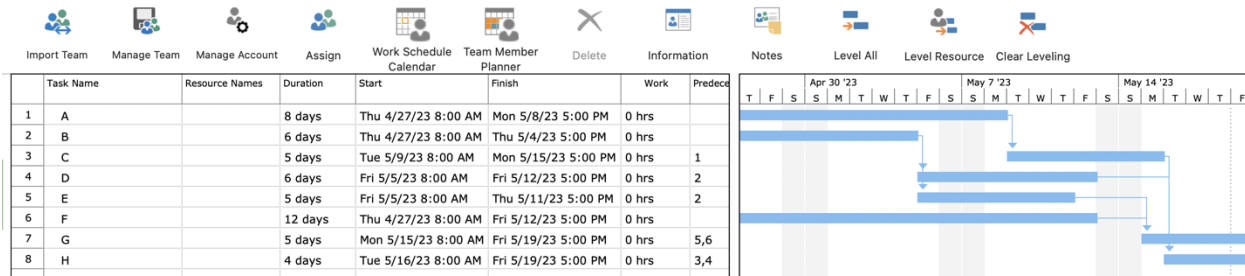
A network diagram is a visual representation of the connections between different devices, nodes, or entities in a network. It can be used to document, plan, and troubleshoot a network infrastructure, and can include network devices, endpoints, and connections between different networks.



(Fig: Network Diagram of given Data)

Gantt Chart:

A Gantt chart is a visual representation of a project schedule that shows the start and end dates of each task, as well as the dependencies between them. It typically includes a horizontal timeline axis and a vertical task axis, with each task represented by a horizontal bar. Gantt charts allow project managers to track progress, allocate resources, and identify potential delays or bottlenecks.



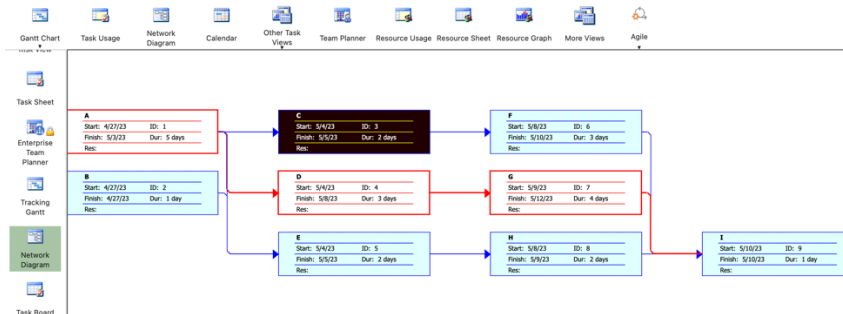
(Fig: Gantt Chart for Given Data)

Lab 2: Draw the precedence network diagram and Gantt Chart using MS Project.

Activity	Duration	Depends on
A	5 days	-
B	1 days	-
C	2 days	A
D	3 days	A
E	2 days	A
F	3 days	C
G	4 days	D
H	2 days	B,E
I	1 days	H
J	1 days	F,G,I

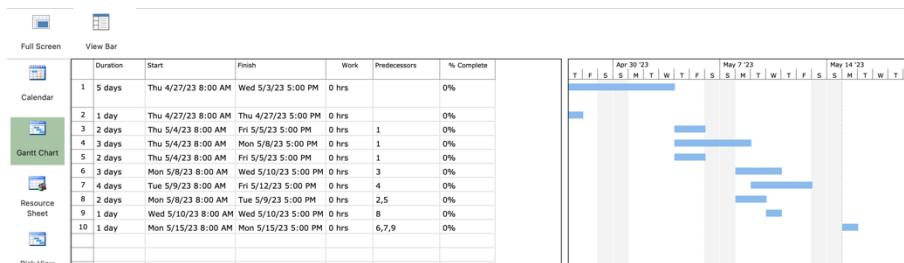
Solution:

Network Diagram:



(Fig: Network Diagram of given Data)

Gantt Chart:



(Fig: Gantt Chart for Given Data)

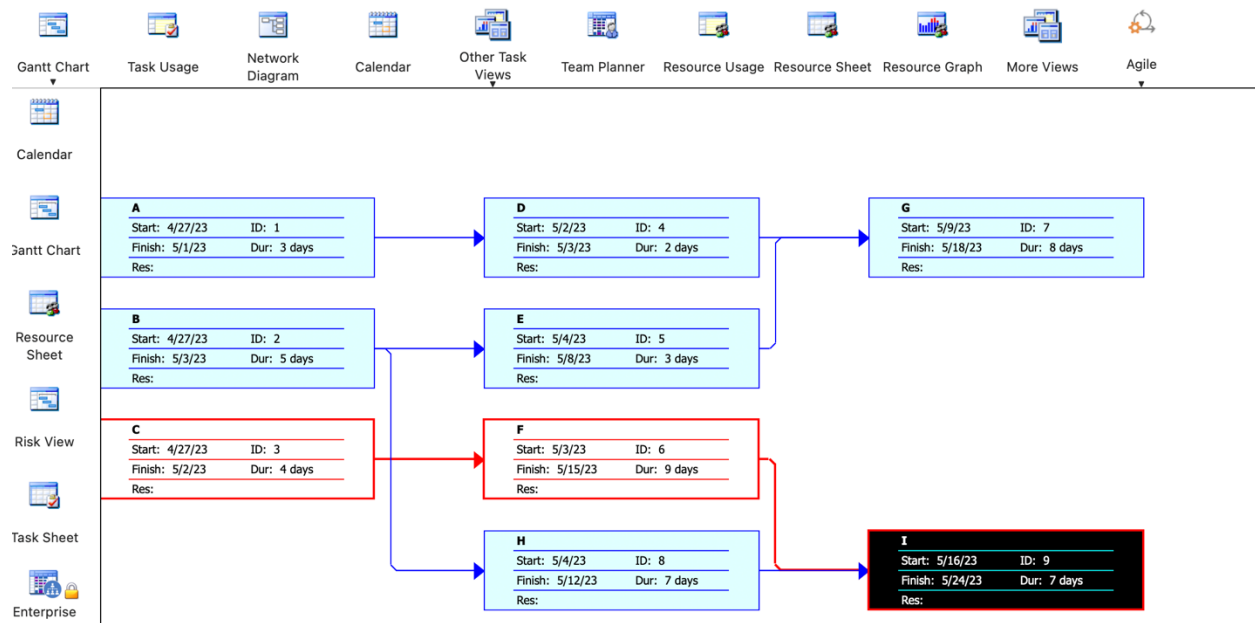
Lab 3: Draw the precedence network diagram and Gantt Chart using MS Project.

(Consider the following table of information:)

Activity	Duration(week)	Precedents
A	3	None
B	5	None
C	4	None
D	2	A
E	3	B
F	9	C
G	8	D,E
H	7	B
I	7	F,H

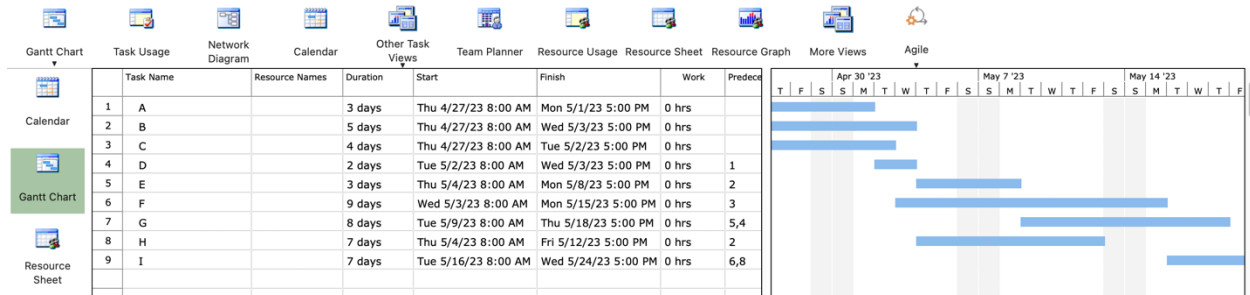
Solution:

Network Diagram:



(Fig: Network Diagram of given Data)

Gantt Chart:



(Fig: Gantt Chart for given data)

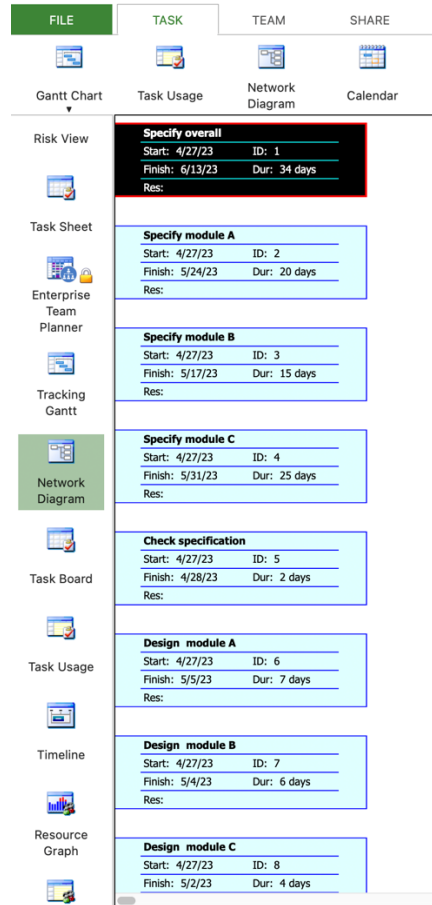
Lab 4: Draw the precedence network diagram and Gantt Chart using MS Project.

(Consider the following table of information:)

Activity	Duration(week)
Specify overall	34
Specify module A	20
Specify module B	15
Specify module C	25
Check specification	2
Design module A	7
Design module B	6
Design module C	4
Code /test A	30
Code /test B	28
Code /test C	15
System Integration	6

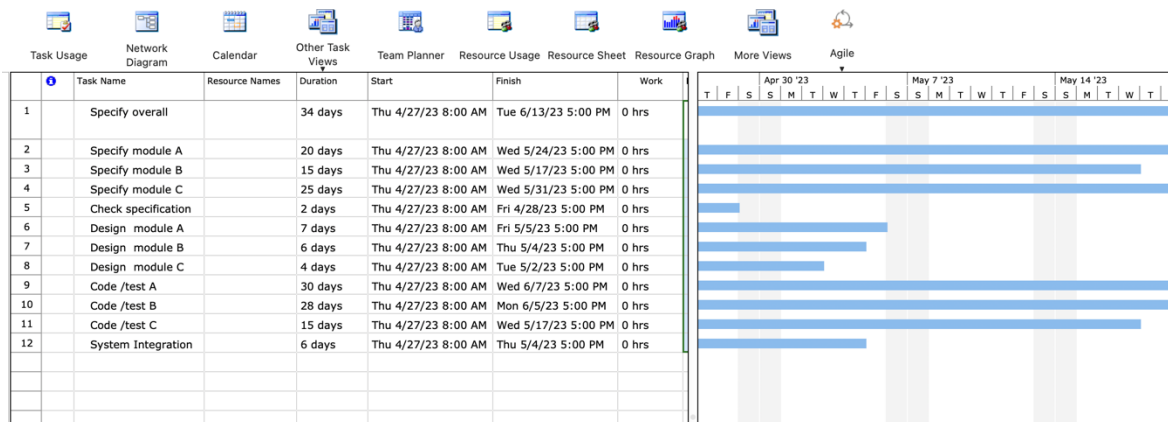
Solution:

Network Diagram:



(Fig: Network Diagram of given Data)

Gantt Chart:



(Fig: Network Diagram of given Data)

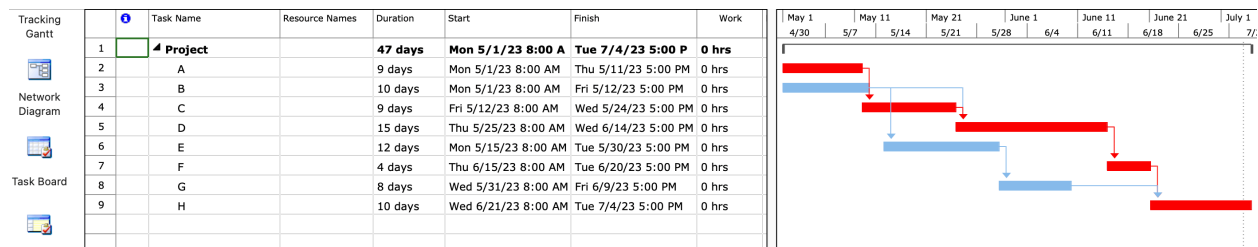
Lab 5: Draw the critical path diagram for the tabulated activities. Identify the critical path and duration of the project.

Activity	Duration	Immediate Predecessor
A	9	-
B	10	-
C	9	A
D	15	B,C
E	12	B
F	4	D
G	8	E
H	10	F,G

Solution:

Critical Path:

The critical path in a project is the sequence of tasks that must be completed in order to meet the project's deadline. It is the longest path of dependent tasks through the project schedule and represents the minimum amount of time required to complete the project. Any delay in a task on the critical path will delay the project's overall completion. Identifying the critical path is important for project managers to allocate resources, prioritize tasks, and manage risks to ensure that the project is completed on time. Here in the figure below, red task indicate the critical path for the project.



(Fig: Critical Path for given data)

Project Duration:

Project duration refers to the amount of time required to complete a project from start to finish. It includes all phases of the project, from initial planning and development to implementation, testing, and deployment. The duration of a project can be affected by a range of factors, including the scope of the project, the number of resources allocated to the project, the complexity of the tasks involved, and unforeseen obstacles or delays

For The above project, the project duration is 47 days

Lab 6: Consider the cash flow of four projects. The cash flow is accounted for at the end of each fiscal year. Calculate ROI and NPV with 10% discount rate for the following data using excel.

<i>Year</i>	<i>Project 1</i>	<i>Project 2</i>	<i>Project 3</i>	<i>Project 4</i>
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net profit	50,000	100,000	50,000	75,000

Solution:

ROI:

ROI stands for Return on Investment, which is a financial metric used to measure the profitability and efficiency of an investment. It is calculated by dividing the net profit generated by an investment by the cost of the investment and expressing the result as a percentage. ROI can be used to evaluate the potential financial benefits of a wide range of investments, including projects, marketing campaigns, new products, and equipment purchases. A high ROI indicates that the investment is profitable and efficient, while a low or negative ROI indicates that the investment is not generating sufficient returns.

$$\text{ROI} = (\text{Net Profit} / \text{Cost of Investment}) \times 100\%$$

NPV:

NPV stands for Net Present Value, which is a financial metric used to determine the profitability of an investment or project. It measures the difference between the present value of cash inflows and the present value of cash outflows over a specified period of time, taking into account the time value of money. The time value of money refers to the fact that a dollar received today is worth more than a dollar received in the future due to the potential for investment and earning interest. A positive NPV indicates that the investment is profitable and will generate returns

above the required rate of return, while a negative NPV indicates that the investment is not profitable and will not generate sufficient returns.

$$NPV = \sum (\text{Cash inflow}_t / (1+r)^t) - \text{Initial investment}$$

Where:

Cash inflows = cash inflow in year t

r = discount rate or required rate of return

t = year of cash flow

Initial investment = total cost of investment at time zero

The ROI and NPV for given data:

Lab 6 - SPM				
Discount Rate	10%			
Year	Project 1	Project 2	Project 3	Project 4
0	-100000	-1000000	-100000	-120000
1	10000	200000	30000	30000
2	10000	200000	30000	30000
3	10000	200000	30000	30000
4	20000	200000	30000	30000
5	100000	300000	30000	75000
Net Profit	50000	100000	50000	75000
NPV	\$620.92	(\$179,750.51)	\$13,723.60	\$21,665.06
Formula for NPV	=NPV(10%,D6:D10)+D5	=NPV(10%,E5:E10)	=NPV(10%,F5:F10)	=NPV(10%,G5:G10)
ROI	50	10	50	62.5
Formula for ROI	=(D11/ABS(D5))*100	=(E11/ABS(E5))*100	=(F11/ABS(F5))*100	=(G11/ABS(G5))*100

(Fig: ROI and NPV using excel)

Lab 7: Perform Earned value analysis of the given project using excel. Calculate ACWP, BCWP, BCWS, SPI, CPI, SV, CV.

Activity	Preceder	Duration (days)	Cost per day	Total cost
A		2	300	600
B	A	3	400	1200
C	B	3	400	1200

D	B	2	200	400
E	D	3	100	300

Field Report at the end of 7 days		
Activity	Actual % Complete	Incurred Cost
A	100	600
B	100	1400
C	33	500
D	50	200
E	0	0

Solution:

Earned Value Analysis (EVA) is a project management technique used to measure the progress and performance of a project against the project plan. It provides a way to compare the planned work to the actual work completed and helps to identify potential problems and delays.

EVA uses three key metrics to measure project progress:

Planned Value (PV): the budgeted cost of the planned work or the budgeted cost of work scheduled (BCWS).

Earned Value (EV): the budgeted cost of the work actually completed or the budgeted cost of work performed (BCWP).

Actual Cost (AC): the actual cost of the work completed or the actual cost of work performed (ACWP).

By comparing these metrics, EVA provides insight into the performance of a project, including:

Schedule Performance Index (SPI): measures the efficiency of the project schedule by comparing the earned value to the planned value. $SPI = EV / PV$.

Cost Performance Index (CPI): measures the efficiency of the project budget by comparing the earned value to the actual cost. $CPI = EV / AC$.

Variance Analysis: compares the planned value to the earned value and the earned value to the actual cost to identify any variances or deviations from the project plan.

The Detail Earn value analysis for the given data:

Gantt Chart:

							Report day			
Activity	day 1	day 2	day 3	day 4	day 5	day 6	day 7	day 8	day 9	day 10
A										
B										
C										
D										
E										

(Fig: Gantt Chart)

Earned Value Analysis:

Activity	total work days on reporting day			
A		2		
B		3		
C		2		
D		2		
E		0		

Activity	ACWP	BCWP	BCWS
A	600	600	600
B	1400	1200	1200
C	500	400	800
D	200	200	400
E	0	0	0
Total	2700	2400	3000

	Value	Formulae
SV	-600	BCWP-BCWS
CV	-300	BCWP-ACWP
SPI	0.8	BCWP/BCWS
CPI	0.88888889	BCWP/ACWP

(Fig: Earned Value Analysis)